Fractal Quantum Entanglement Density via Fibonacci Embedding in FBS³R

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Abstract

We formulate a fractal cosmological model (FBS³R) in which spatial slices are compact 3-spheres S^3 and the discrete node hierarchy evolves by a minimal positive second-order recursion. Via the Perron–Frobenius theorem we prove that the golden ratio $\varphi = (1+\sqrt{5})/2$ is the unique dominant eigenvalue of this recursion—not a free parameter. A Spectral–Geometric Consistency postulate (H3) bridges the combinatorial and geometric sectors by equating the maximal Laplacian mode index on S^3 to the node count, thereby deriving $R_n \propto \varphi^n$. Scaling laws $C_n \to \varphi^{-2}$, $I_n \sim \kappa\varphi^{2n}$, and $D_n \sim \varphi^{-n}$ are established as theorems. The φ-exponents are shown to be robust under arbitrary connectivity schemes. Entropy is excluded from the core derivations absent a rigorous degree-distribution model.

0. Reader's Map: What Is Proved vs. Assumed

Every statement in this paper carries one of the following explicit epistemic labels:

- **Postulate / Assumption** — a structural condition defining the model (an input, not a claim about the world).
- **Definition** — precise designation of a quantity.
- **Lemma / Theorem** — a strict mathematical consequence of the postulates, with proof.
- **Remark** — commentary on physical interpretation; not part of any proof.

The principal objective is to show that φ is not a tuning parameter but the dominant eigenvalue of the minimal non-degenerate discrete composition of levels, and that the scaling laws for correlation and informational density follow as theorems.

I. Geometric Sector

Postulate G (Compact Boundaryless Spatial Topology)

Spatial slices are assumed to be compact, connected, boundaryless 3-manifolds. In the spherical sector:

$$\Sigma \simeq S^3 .$$

Remark. This is a geometric-sector postulate only. No dynamical equations are introduced at this stage. The choice of S^3 is a structural Ansatz; alternative compact topologies (lens spaces L(p,q)) are noted in Section XII.

II. Discrete Hierarchical Sector

Postulate H1 (Discrete Hierarchy)

There exists a discrete hierarchy of levels indexed by $n \in \mathbb{N}$. At each level:

- $F_n \in \mathbb{N}$ — number of elementary structural nodes.
- $R_n > 0$ — characteristic geometric scale.

Postulate H2 (Minimal Non-Degenerate Second-Order Recursion)

The node count F_n satisfies a positive integer recursion of second order. In the minimal case (coefficients a = b = 1):

$$F_{n+1} = F_n + F_{n-1}, \qquad F_1 \geq 1, \quad F_2 \geq 1.$$

Remark. Second order reflects coupling between consecutive levels. Minimality (a = b = 1) is the unique choice minimising the dominant growth rate in the class of positive integer second-order recursions (Theorem 6). The golden ratio φ is a consequence of this postulate, not a premise.

III. Spectral Formulation and Perron–Frobenius Control

Definition 1 (State Vector and Transfer Matrix)

Define the state vector $v_n = (F_n, F_{n-1})^T$ and transfer matrix:

$$T = [[1, 1], [1, 0]], \qquad \text{so that} \quad v_{n+1} = T\, v_n.$$

Lemma 1 (Eigenvalues of the Fibonacci Transfer Matrix)

The characteristic polynomial of T is $\lambda^2 - \lambda - 1 = 0$, with eigenvalues:

$$\lambda_\pm = (1 \pm \sqrt{5}) / 2.$$

Proof. Standard computation from the quadratic characteristic equation. □

Theorem 1 (Dominant Eigenvalue and Asymptotic Ratio)

For any positive initial conditions $F_1, F_2 > 0$:

$$\lim_{n\to\infty} F_{n+1} / F_n = \varphi = (1 + \sqrt{5})/2 \approx 1.6180.$$

Proof. T is non-negative and primitive ($T^2 = [[2,1],[1,1]]$ is strictly positive). By the Perron–Frobenius theorem, T has a unique positive dominant eigenvalue equal to its spectral radius $\rho(T) = \lambda_+ = \varphi$, with a strictly positive eigenvector. For any positive initial vector v_2, the iterates $T^n v_2$ align asymptotically with this eigenvector, so consecutive component ratios converge to φ.

Remark. Theorem 1 removes φ from the list of model inputs. It is the forced dominant exponent of the minimal positive second-order recursion.

IV. Robustness: φ Under Perturbations

Postulate H2' (General Positive Second-Order Recursion)

Consider the broader class:

$$F_{n+1} = a\, F_n + b\, F_{n-1}, \qquad a, b \in \mathbb{N}, \quad a,b \geq 1.$$

Lemma 2 (Dominant Eigenvalue for General Positive Recursion)

The dominant eigenvalue of the corresponding transfer matrix is:

$$\lambda_+(a, b) = (a + \sqrt{(a^2 + 4b)}) / 2.$$

Proof. Direct application of the quadratic formula to the characteristic polynomial $\lambda^2 - a\lambda - b = 0$.

Theorem 2 (Structural Stability of Dominant Scaling)

For any $a, b \geq 1$ and positive initial conditions: $\lim_{n\to\infty} F_{n+1}/F_n = \lambda_+(a,b)$.

Proof. The transfer matrix $T_{a,b}$ is primitive for all $a, b \geq 1$. The Perron–Frobenius argument of Theorem 1 applies verbatim.

Corollary 1 (Minimality of φ in the Positive Integer Class)

$\lambda_+(a, b)$ is strictly increasing in both a and b. The minimum over $a, b \in \mathbb{N}_{\geq 1}$ is attained uniquely at $(a, b) = (1,1)$:

$$\lambda_+(1, 1) = \varphi.$$

Remark. φ is not a convenient choice but the slowest-growing dominant exponent in the entire class of positive integer second-order recursions.

V. Spectral–Geometric Consistency and Derivation of R_n

Postulate H3 (Spectral–Geometric Consistency)

The Laplacian spectrum on $S^3(R_n)$ is discrete:

$$\lambda_\ell = \ell(\ell+2) / R_n^2, \qquad \ell \in \mathbb{N}.$$

We assume the maximal structurally realisable mode index equals the node count:

$$\ell_{max}(n) \sim F_n.$$

Requiring invariance of modal density across levels:

$$\ell_{max}(n) / R_n = const \quad \Rightarrow \quad F_n / R_n = const \quad \Rightarrow \quad R_n \propto F_n \sim \varphi^n.$$

$$Therefore: \quad R_n = R_0 \varphi^n.$$

Remark. H3 is the bridge between the combinatorial and geometric sectors. It asserts that the number of available Laplacian modes on Σ_n matches the number of structural degrees of freedom at level n. Under this consistency condition R_n inherits the φ-exponential from the node-count asymptotics rather than being separately postulated.

VI. Correlation Density

Definition 2 (Node Increment and Correlation Density)

Define $\Delta F_n := F_n - F_{n-1}$. In the Fibonacci case, $\Delta F_n = F_{n-2}$. The correlation fraction:

$$C_n := \Delta F_n / F_n.$$

Theorem 3 (Asymptotic Correlation Density)

In the Fibonacci model:

$$\lim_{n\to\infty} C_n = \varphi^{-2} \approx 0.382.$$

Proof. $C_n = F_{n-2} / F_n = (F_{n-2}/F_{n-1}) \cdot (F_{n-1}/F_n) \to \varphi^{-1} \cdot \varphi^{-1} = \varphi^{-2}$. □

Remark. The corrected limit is $\varphi^{-2} \approx 0.382$, superseding the preliminary value of $\varphi^{-1} \approx 0.618$, which resulted from the incorrect identification $\Delta F_n = F_{n-1}$ rather than F_{n-2}.

VII. Connectivity, Correlations, and Robustness

Postulate C1 (Minimal Hierarchical Attachment)

At step n, new nodes attach to existing nodes subject to: (i) connections arise only between levels n and n−1; (ii) no rewiring of prior connections occurs. With coupling density $\kappa \in (0,1]$:

$$I_n := \kappa\, F_{n-1}\, \Delta F_n.$$

Remark. C1 is the minimal hierarchical scheme, not a physical assertion. $\kappa = 1$ gives full bipartite coupling; $\kappa < 1$ sparse coupling. Theorem 4 shows the φ-exponent is κ-independent.

Theorem 4 (Scaling of New Correlations)

For any $\kappa > 0$ in the Fibonacci model: $I_n \sim (\kappa/5)\, \varphi^{2n-3}$. In particular the growth exponent 2 (base φ) is independent of κ.

Proof. With $\Delta F_n = F_{n-2}$ and Binet's formula $F_n \sim \varphi^n/\sqrt{5}$: $I_n = \kappa\, F_{n-1}\, F_{n-2} \sim \kappa(\varphi^{n-1}/\sqrt{5})(\varphi^{n-2}/\sqrt{5}) = (\kappa/5)\varphi^{2n-3}$. The exponent 2n is fixed; κ affects only the prefactor.

Remark (Generality of the φ^{2n} Exponent)

Under random-graph, small-world, or scale-free connectivity with a linear number of new edges per step, the growth exponent of I_n is determined by the product $F_n \cdot F_{n-1} \sim \varphi^{2n}$. The φ-exponents are therefore a robust consequence of the dominant eigenvalue structure, not of the specific connectivity rule.

VIII. Informational Density

Definition 3 (Combinatorial Informational Density)

The informational density of new correlations is $D_n := I_n / V_n$, where $V_n \propto R_n^3$ is the 3-volume at scale R_n.

Theorem 5 (Decay of Informational Density)

In the Fibonacci-minimal model:

$$D_n \propto \varphi^{-n} \qquad (n \to \infty).$$

Proof. $I_n \sim \varphi^{2n}$ (Theorem 4) and $V_n \propto R_n^3 = R_0^3\, \varphi^{3n}$ (Postulate H3). Therefore $D_n \sim \varphi^{2n} / \varphi^{3n} = \varphi^{-n}$.

Remark. This is a purely combinatorial and geometric result. No thermodynamic or quantum-mechanical entropy is invoked.

IX. Uniqueness of φ in the Minimal Positive Class

Theorem 6 (Uniqueness of φ as Minimum Dominant Eigenvalue)

Among all recursions $F_{n+1} = aF_n + bF_{n-1}$ with a, b ∈ $\mathbb{N}_{\geq 1}$:

$$\min_{a,b} \lambda_+(a,b) = \lambda_+(1,1) = \varphi.$$

Proof. $\lambda_+(a,b) = (a + \sqrt{(a^2+4b)})/2$ has positive partial derivatives in a and b for all a, b > 0. The lattice minimum over a, b ∈ $\mathbb{N}_{\geq 1}$ is therefore attained uniquely at (a,b) = (1,1), giving $\lambda_+ = \varphi$.

Remark. φ is the minimal expansion rate compatible with a non-degenerate positive integer hierarchical recursion. It is forced by the model class, not chosen.

X. On Entropy: Scope and Exclusion

Earlier versions of this work included the claim $S_n \sim n \log \varphi$ for entanglement entropy. We exclude this from the present paper. Entanglement entropy in a graph-theoretic setting requires a well-defined probability distribution over the degree sequence. If all new nodes at level n acquire the same degree ~ F_{n-1}, the degree distribution is degenerate and the Shannon entropy is zero. A non-trivial entropy requires either (a) a probabilistic attachment rule, or (b) an explicit bipartite quantum state assignment. Both extensions are deferred to future work.

XI. Summary and Conclusions

The table below summarises the principal results.

Quantity	Scaling Law	Status
Correlation density C_n	$C_n \to \varphi^{-2} \approx 0.382$	Theorem 3
New correlations I_n	$I_n \sim \kappa\varphi^{2n}$	Theorem 4
Informational density D_n	$D_n \propto \varphi^{-n}$	Theorem 5
Geometric scale R_n	$R_n = R_0\varphi^n$	H3 (derived from Laplacian)
Dominant eigenvalue	$\lambda_+ = \varphi$	Theorem 1 (Perron–Frobenius)
Entropy S_n	Deferred	Excluded — see Section X

The FBS³R framework, as formalised here, establishes that:

- φ is forced as the dominant exponent by the minimal positive second-order recursion, via Perron–Frobenius.
- The geometric scale R_n follows from a spectral–geometric consistency condition (H3), not from a separate postulate.
- Correlation and informational density scale as φ^{2n} and φ^{-n} respectively, as strict mathematical consequences of the postulates.
- The φ-exponents are robust under arbitrary connectivity schemes (κ-independence).
- All metaphorical language (Indra's Net, self-aware lattice) is excluded from the mathematical core.

XII. Limitations and Future Directions

1. **Entropy extension.** Incorporate a probabilistic attachment rule to yield a non-trivial degree distribution and enable a rigorous Shannon or von Neumann entropy derivation.
2. **Dynamical sector.** Derive H3 from a variational principle or link R_n to a Laplacian-based Hamiltonian on S^3, converting the consistency postulate into a theorem.
3. **Topology generalisation.** Extend beyond S^3 to lens spaces L(p,q) and investigate how the modified Laplacian spectrum alters H3.
4. **Observational connection.** Identify candidate observational signatures of φ-scaling in the CMB power spectrum or large-scale structure statistics.
5. **Higher-order recursions.** Study third- and higher-order positive recursions and characterise their dominant eigenvalue structure by analogy with Theorems 1–2.

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